

# A QUANTUM JUMP



**Quantum computing** uses the laws of quantum physics to solve certain complex problems exponentially faster than classical computers



**The technology** could reshape drug discovery, climate modelling, logistics, finance, AI, and materials science



**Experts estimate** Q-Day could arrive around 2029-30, when quantum systems may begin breaking today's encryption standards



**India launched** the ₹6,003 crore National Quantum Mission in 2023



**It aims to build capabilities** in quantum computing, communication, sensing, and advanced quantum hardware by 2031



**India still trails global leaders** such as the US and China in quantum hardware development



**India's strengths** in mathematics, engineering, and software could give it a lead in quantum algorithms, applications, and cybersecurity



**Andhra Pradesh is building a Quantum Valley.** Karnataka has unveiled a ₹1,000 crore state quantum mission and Quantum Hardware Park plans

## GLOBAL QUANTUM COMPUTING MARKET SIZE

**\$3.5**

BILLION IN 2025

**\$20**

BILLION (PROJECTED) IN 2030

**SOURCE** MARKETS & MARKETS RESEARCH



**FOR DECADES**, we've been told that the secret to digital safety is complexity. Add uppercase letters. Throw in numbers. Sprinkle special characters like #, %, and &. The more random the password looks, the safer your bank account, email, and identity supposedly become.

In classical computing, this logic has held firm. Modern encryption is protected by mathematics so vast that even the world's fastest supercomputers would need billions, sometimes trillions, of years to crack it. Our financial systems, military communications, cryptocurrencies, and even WhatsApp messages rest on one assumption: that the math is too hard and the clock will run out before hackers ever succeed.

But the foundations of that digital trust are beginning to shift.

An emerging generation of machines—quantum computers—does not play by the rules of conventional computing. If a traditional computer is like a locksmith testing one key at a time, a quantum computer can explore countless possibilities simultaneously. Problems that would take today's most powerful computers thousands of years could, in theory, be solved in hours or minutes. In that world, the strength of passwords may no longer matter.

The fear surrounding Q-Day—the expected moment when quantum computers become powerful enough to crack encryption algorithms that secure the internet—is one facet of the story. Quantum computing represents a much larger technological shift, one that could reshape industries ranging from healthcare and materials science to finance and energy. Here is how it works: Traditional computers process information step by step using bits that exist as either 0 or 1. Quantum computers use qubits, which can exist in multiple states simultaneously, allowing them to process many possibilities at once instead of checking every option sequentially. Experts say this could help in areas such as drug discovery, battery design, logistics optimisation, climate modelling, and advanced financial simulations, tasks where the number of variables overwhelms conventional systems.

This has triggered an intense global push towards mastering the technology. IBM, Google, Microsoft, and Intel are racing to build stable quantum systems, while governments, from the United States to China, are treating quantum capabilities as strategic assets comparable to artificial intelligence or advanced semiconductor infrastructure. Having missed out on semiconductor manufacturing and artificial intelligence revolutions, India has launched a National Quantum